

Homework 7

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Problem 1

Summary

For the series Y_t I chose a MA(2) model with exogenous regressor X_{t+3} .

Coefficients:

	Estimate	SE	t.value	p.value
ma1	0.1878	0.0347	5.4136	0.0000
ma2	0.5652	0.0329	17.1608	0.0000
intercept	-6.5233	5.9041	-1.1049	0.2697
xreg	1.2813	0.0293	43.6641	0.0000

Explanation

Both X_t and Y_t are stationary and do not have changing variance over time or seasonality. The ACF and PACF show 3 significant lags in the ACF and 2 significant lags in the PACF, so models with a maximum p of 2 and a maximum q of 3 were considered. The AR(2) model had the best AICc and BIC of these models and good diagnostics apart from some of the later values of the LBP. The intercept was included because it was significant at all levels.

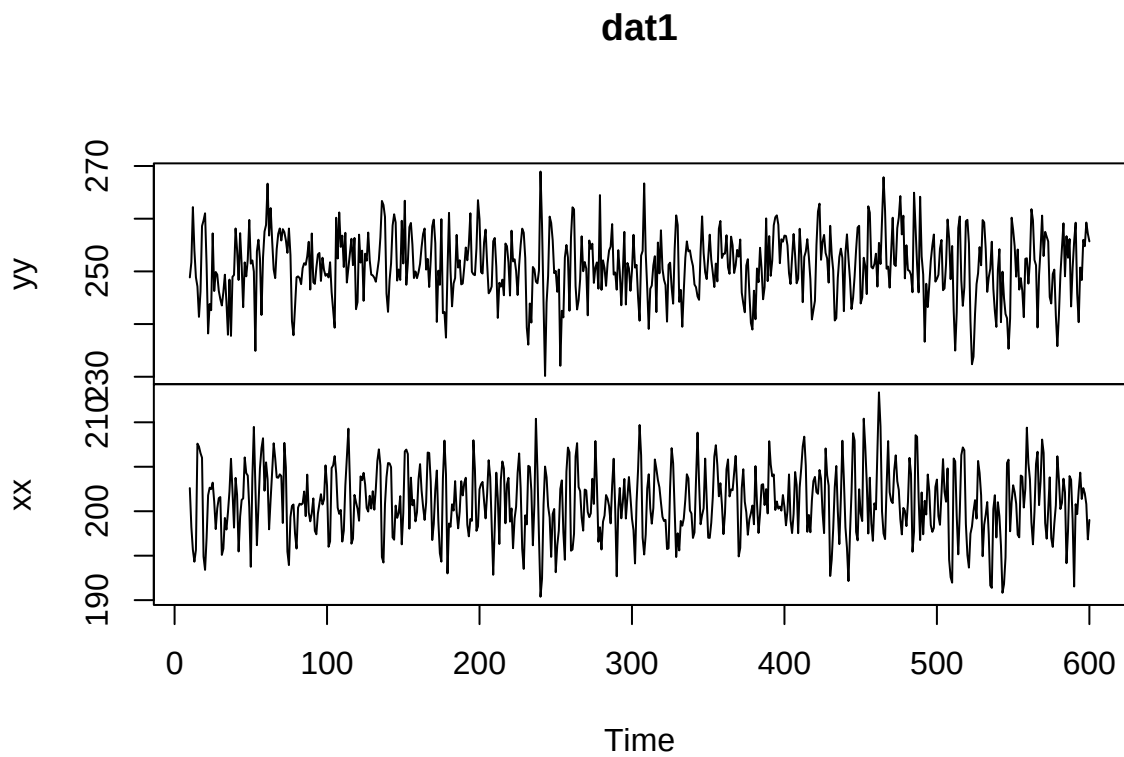
After prewhiteneing both X_t and Y_t with this model, the ccf shows a significant correlation at -3, indicating that X_t leads Y_t by 3. Thus X_{t+3} was used in the design matrix with Y_t .

For the model of Y_t including the exogenous regressor X_{t+3} , the ACF and PACF show 3 significant lags in the ACF and 4 significant lags in the PACF, so models with a maximum p of 4 and a maximum q of 3 were considered. Of these, the MA(2) had the best AICc and BIC and strong diagnostic plots, so it was the chosen model.

Output

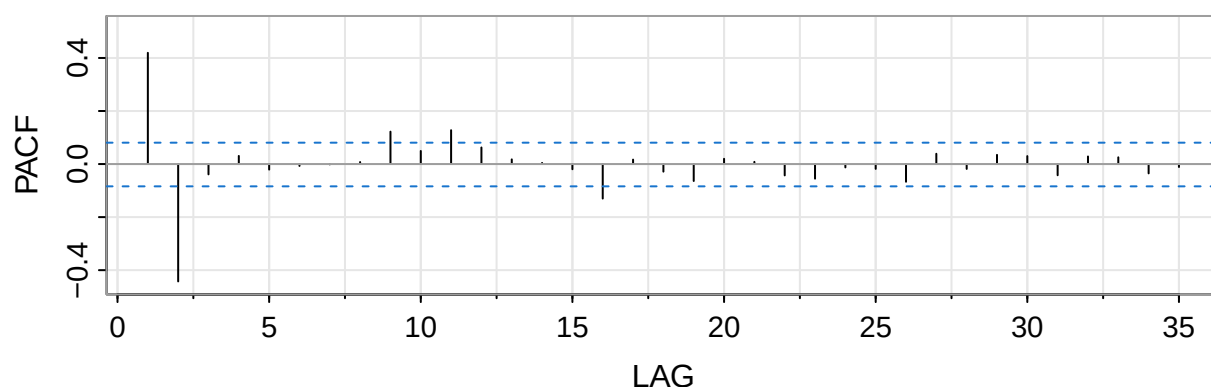
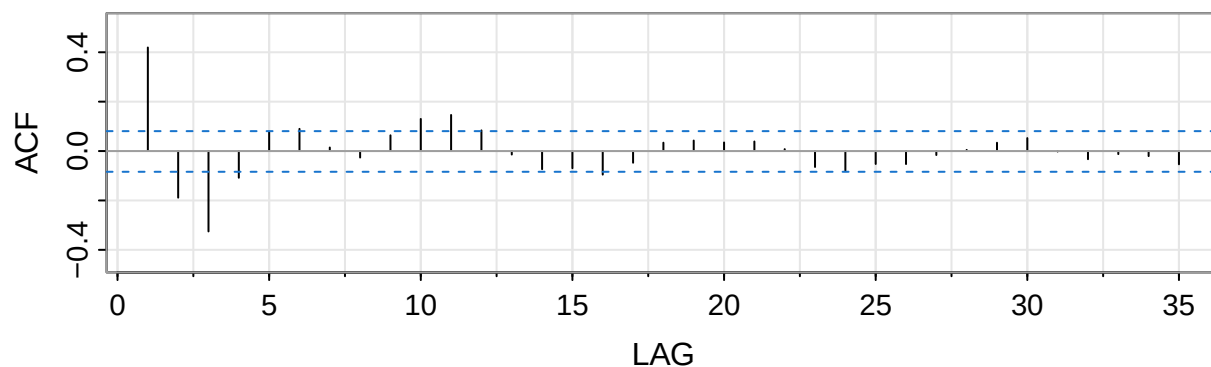
```
library(astsa)
load("hw7.rsav")
# Access the 'xx' column
xx <- dat1[, "xx"]
```

```
plot(dat1)
```



```
# Compute and plot the ACF and PACF of 'xx'  
acf2(xx, main = "ACF and PACF of xx")
```

ACF and PACF of xx



```
##      [,1] [,2] [,3] [,4] [,5] [,6] [,7] [,8] [,9] [,10] [,11] [,12] [,13]
## ACF  0.42 -0.19 -0.33 -0.11  0.08  0.09  0.01 -0.03  0.06  0.13  0.15  0.08 -0.01
## PACF  0.42 -0.44 -0.04  0.03 -0.02 -0.01  0.00  0.01  0.12  0.05  0.13  0.06  0.02
##      [,14] [,15] [,16] [,17] [,18] [,19] [,20] [,21] [,22] [,23] [,24] [,25]
## ACF  -0.07 -0.07 -0.10 -0.05  0.03  0.04  0.03  0.04  0.01 -0.06 -0.08 -0.05
## PACF  0.00 -0.02 -0.13  0.02 -0.03 -0.06  0.02  0.01 -0.04 -0.06 -0.01 -0.02
##      [,26] [,27] [,28] [,29] [,30] [,31] [,32] [,33] [,34] [,35]
## ACF  -0.05 -0.02  0.00  0.03  0.05  0.00 -0.03 -0.01 -0.02 -0.05
## PACF -0.07  0.04 -0.02  0.03  0.03 -0.04  0.03  0.03 -0.04 -0.01

maxAR<-2+1
maxMA<-3+1
fits_all<-vector("list",length=maxAR)
AICmin<-BICmin<-Inf
for (ii in 1:maxAR){
  fits_all[[ii]]<-vector("list",length=maxMA)
  for (jj in 1:maxMA){
    fits_all[[ii]][[jj]]<-invisible(sarima(xx,p=ii-1,d=0,q=jj-1,no.constant=FALSE, details=FALSE))
    if(fits_all[[ii]][[jj]]$ICs["AICc"]<AICmin) AICmin<-fits_all[[ii]][[jj]]$ICs["AICc"]
    if(fits_all[[ii]][[jj]]$ICs["BIC"]<BICmin) BICmin<-fits_all[[ii]][[jj]]$ICs["BIC"]
  }
}

## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE  t.value p.value
```

```

## xmean 201.0404 0.1565 1284.757      0
##
## sigma^2 estimated as 14.47146 on 590 degrees of freedom
##
## AIC = 5.516824  AICc = 5.516835  BIC = 5.531652
##
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE  t.value p.value
## ma1      0.5700 0.0300  18.9769      0
## xmean 201.0442 0.2096 959.0402      0
##
## sigma^2 estimated as 10.54956 on 589 degrees of freedom
##
## AIC = 5.204779  AICc = 5.204813  BIC = 5.227021
##
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE  t.value p.value
## ma1      0.6749 0.0486  13.8776  0.000
## ma2      0.1544 0.0560   2.7579  0.006
## xmean 201.0435 0.2428 828.0842  0.000
##
## sigma^2 estimated as 10.428 on 588 degrees of freedom
##
## AIC = 5.196697  AICc = 5.196766  BIC = 5.226354
##
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE  t.value p.value
## ma1      0.5509 0.0407  13.5427  0.0000
## ma2     -0.0717 0.0510  -1.4063  0.1602
## ma3     -0.2628 0.0410  -6.4081  0.0000
## xmean 201.0469 0.1571 1279.9357  0.0000
##
## sigma^2 estimated as 9.843113 on 587 degrees of freedom
##
## AIC = 5.142511  AICc = 5.142627  BIC = 5.179583
##
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE  t.value p.value
## ar1      0.4189 0.0373  11.2333      0
## xmean 201.0399 0.2442 823.4000      0
##
## sigma^2 estimated as 11.9248 on 589 degrees of freedom
##
## AIC = 5.326976  AICc = 5.32701  BIC = 5.349219
##
## <><><><><><><><><><><><><><>

```

```

##
## Coefficients:
##      Estimate      SE  t.value p.value
## ar1      0.0958 0.0610   1.5704  0.1169
## ma1      0.5147 0.0479  10.7427  0.0000
## xmean 201.0428 0.2232 900.7508  0.0000
##
## sigma^2 estimated as 10.50736 on 588 degrees of freedom
##
## AIC = 5.204189  AICc = 5.204258  BIC = 5.233846
##
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE  t.value p.value
## ar1      0.6513 0.0935   6.9668  0.0000
## ma1     -0.1484 0.0865  -1.7154  0.0868
## ma2     -0.4495 0.0477  -9.4225  0.0000
## xmean 201.0458 0.1531 1312.9898  0.0000
##
## sigma^2 estimated as 10.39673 on 587 degrees of freedom
##
## AIC = 5.196968  AICc = 5.197083  BIC = 5.234039
##
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE  t.value p.value
## ar1      0.3051 0.1017   2.9997  0.0028
## ma1      0.2786 0.0959   2.9063  0.0038
## ma2     -0.2226 0.0595  -3.7416  0.0002
## ma3     -0.2754 0.0379  -7.2747  0.0000
## xmean 201.0484 0.1443 1393.7099  0.0000
##
## sigma^2 estimated as 9.716534 on 586 degrees of freedom
##
## AIC = 5.132985  AICc = 5.133158  BIC = 5.17747
##
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE  t.value p.value
## ar1      0.6049 0.0369  16.4136    0
## ar2     -0.4428 0.0369 -12.0096    0
## xmean 201.0472 0.1520 1322.3012    0
##
## sigma^2 estimated as 9.580412 on 588 degrees of freedom
##
## AIC = 5.1122  AICc = 5.112269  BIC = 5.141857
##
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE  t.value p.value

```

```

## ar1      0.6690 0.0779      8.5910 0.0000
## ar2     -0.4698 0.0452    -10.4034 0.0000
## ma1     -0.0798 0.0871     -0.9157 0.3602
## xmean 201.0467 0.1463 1374.1238 0.0000
##
## sigma^2 estimated as 9.566829 on 587 degrees of freedom
##
## AIC = 5.114172  AICc = 5.114287  BIC = 5.151243
##
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE    t.value p.value
## ar1      0.6205 0.1066     5.8235 0.0000
## ar2     -0.4936 0.0553    -8.9180 0.0000
## ma1     -0.0309 0.1161    -0.2658 0.7905
## ma2      0.0670 0.0915     0.7317 0.4646
## xmean 201.0462 0.1510 1331.3160 0.0000
##
## sigma^2 estimated as 9.558179 on 586 degrees of freedom
##
## AIC = 5.116657  AICc = 5.11683  BIC = 5.161142
##
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE    t.value p.value
## ar1      0.5905 0.1182     4.9937 0.0000
## ar2     -0.4365 0.1162    -3.7555 0.0002
## ma1     -0.0013 0.1241    -0.0107 0.9915
## ma2      0.0276 0.1063     0.2596 0.7952
## ma3     -0.0476 0.0814    -0.5840 0.5595
## xmean 201.0470 0.1472 1365.8533 0.0000
##
## sigma^2 estimated as 9.552732 on 585 degrees of freedom
##
## AIC = 5.119475  AICc = 5.119719  BIC = 5.171375
##

```

```

nn <- length(xx)
#look at AICs and BICs
print("check AICcs")

```

```

## [1] "check AICcs"

```

```

## [1] "check AICcs"
for (ii in 1:maxAR){
  for(jj in 1:maxMA){
    print(paste0(ii-1,"",jj-1))
    print(((fits_all[[ii]][[jj]])$ICs["AICc"] - AICmin) * nn)
  }
}

```

```

## [1] "00"
##      AICc

```

```

## 239.0987
## [1] "01"
##      AICc
## 54.6937
## [1] "02"
##      AICc
## 49.93804
## [1] "03"
##      AICc
## 17.94168
## [1] "10"
##      AICc
## 126.9123
## [1] "11"
##      AICc
## 54.36552
## [1] "12"
##      AICc
## 50.12545
## [1] "13"
##      AICc
## 12.3458
## [1] "20"
## AICc
##      0
## [1] "21"
##      AICc
## 1.193013
## [1] "22"
##      AICc
## 2.695938
## [1] "23"
##      AICc
## 4.402948

```

```
print("check BICs")
```

```
## [1] "check BICs"
```

```

## [1] "check BICs"
for (ii in 1:maxAR){
for(jj in 1:maxMA){
print(paste0(ii-1,"",jj-1))
print(((fits_all[[ii]][[jj]])$ICs["BIC"] - BICmin) * nn)
}
}

```

```

## [1] "00"
##      BIC
## 230.3692
## [1] "01"
##      BIC
## 50.33236
## [1] "02"
##      BIC

```

```

## 49.93804
## [1] "03"
##      BIC
## 22.29612
## [1] "10"
##      BIC
## 122.551
## [1] "11"
##      BIC
## 54.36552
## [1] "12"
##      BIC
## 54.47989
## [1] "13"
##      BIC
## 21.04775
## [1] "20"
## BIC
##      0
## [1] "21"
##      BIC
## 5.547456
## [1] "22"
##      BIC
## 11.39789
## [1] "23"
##      BIC
## 17.44545

sarima(xx,p=2,d=0,q=0,no.constant=FALSE, details=TRUE)

```

```

## initial  value 1.337174
## iter    2 value 1.173614
## iter    3 value 1.145858
## iter    4 value 1.130910
## iter    5 value 1.130552
## iter    6 value 1.130551
## iter    7 value 1.130551
## iter    8 value 1.130550
## iter    9 value 1.130550
## iter   10 value 1.130550
## iter   11 value 1.130550
## iter   11 value 1.130550
## iter   11 value 1.130550
## final   value 1.130550
## converged
## initial  value 1.130394
## iter    2 value 1.130394
## iter    3 value 1.130393
## iter    3 value 1.130393
## iter    3 value 1.130393
## final   value 1.130393
## converged
## <><><><><><><><><><><><><><>
##

```


The figure displays three diagnostic plots for the fitted model (2,0,0):

- Standardized Residuals:** A time series plot showing the standardized residuals over 600 time units. The residuals fluctuate around zero, indicating a good fit.
- ACF of Residuals:** A plot of the Autocorrelation Function (ACF) for the residuals. The ACF values are mostly within the 95% confidence interval (dashed blue lines), suggesting the residuals are uncorrelated.
- Normal Q-Q Plot of Std Residuals:** A plot comparing the sample quantiles of the standardized residuals against the theoretical quantiles of a normal distribution. The points closely follow the diagonal line, indicating approximate normality.

```
## <><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE    t.value p.value
## ar1       0.6049 0.0369   16.4136     0
## ar2      -0.4428 0.0369  -12.0096     0
## xmean    201.0472 0.1520 1322.3012     0
##
## sigma^2 estimated as 9.580412 on 588 degrees of freedom
##
## AIC = 5.1122  AICc = 5.112269  BIC = 5.141857
##
```

```

# Pre-whiten xx: Extract residuals from the sarima fit
xx_prewhitened <- sarima_fit$fit$residuals

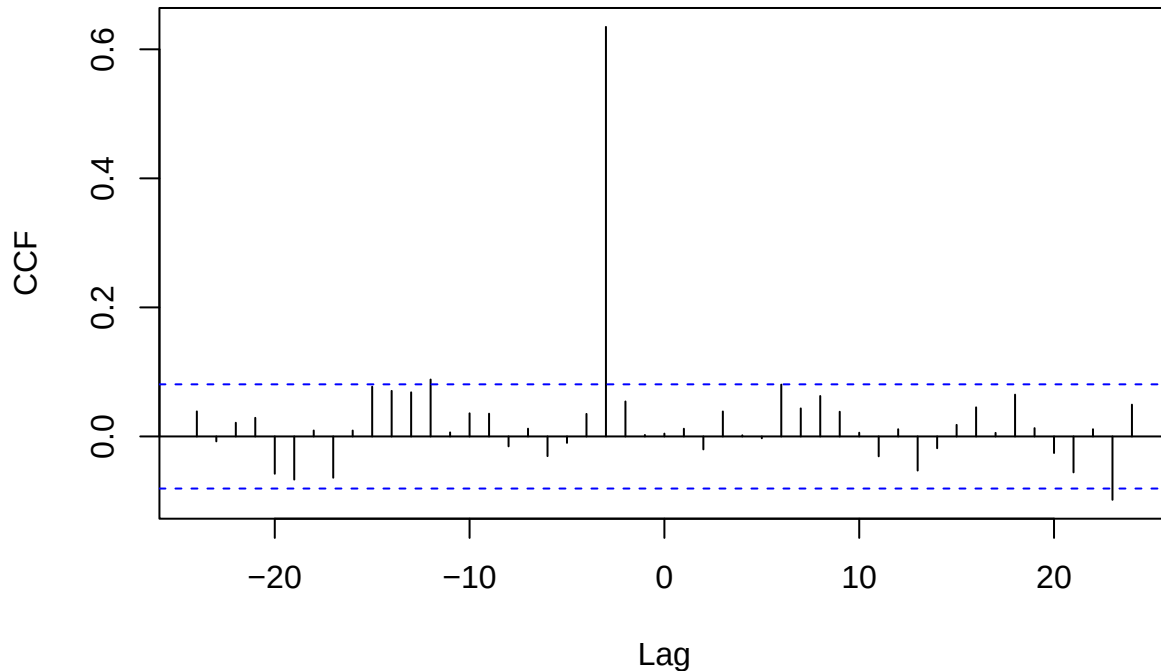
# Get AR(2) coefficients and intercept
ar_coeffs <- coef(sarima_fit$fit)[1:2] # AR(2) coefficients
intercept <- coef(sarima_fit$fit)["intercept"]

# Pre-whiten yy using the AR(2) filter derived from xx
yy <- dat1[, "yy"]
yy_prewhitened <- stats::filter(yy, filter = c(1, -ar_coeffs), method = "convolution", sides = 1)

# Plot the cross-correlation function (CCF) of the pre-whitened series
ccf(xx_prewhitened, yy_prewhitened, main = "CCF of Pre-whitened xx and yy", ylab = "CCF", na.action = na)

```

CCF of Pre-whitened xx and yy



Design matrix created using the lag -3 of x based on CCF

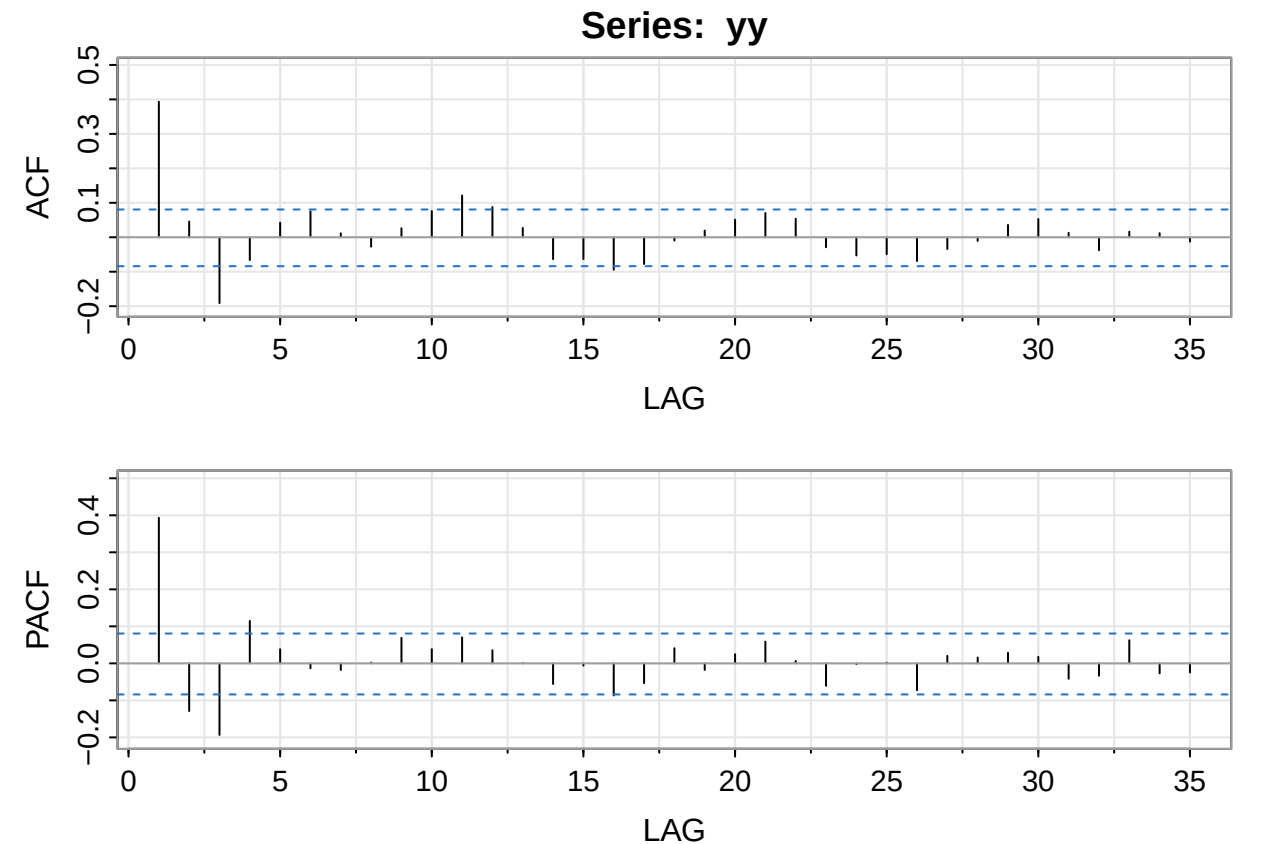
```

# Create lagged values for xx at lag 3
xx_lag3 <- stats::lag(dat1[, "xx"], -3)

# Combine yy and the lagged xx values into a time-series object
design_matrix <- ts.intersect(yy = dat1[, "yy"], xx_lag3 = xx_lag3)
# Remove rows with NA values caused by lagging
design_matrix <- na.omit(design_matrix)

```

```
acf2(yy)
```



```
##      [,1] [,2] [,3] [,4] [,5] [,6] [,7] [,8] [,9] [,10] [,11] [,12] [,13]
## ACF  0.39  0.05 -0.19 -0.07  0.04   0.08  0.01 -0.03  0.03   0.08  0.12  0.09  0.03
## PACF 0.39 -0.13 -0.19  0.11  0.04 -0.01 -0.02  0.00  0.07   0.04  0.07  0.04  0.00
##      [,14] [,15] [,16] [,17] [,18] [,19] [,20] [,21] [,22] [,23] [,24] [,25]
## ACF -0.06 -0.06 -0.09 -0.08 -0.01  0.02  0.05  0.07  0.05 -0.03 -0.05 -0.05
## PACF -0.06 -0.01 -0.09 -0.05  0.04 -0.02  0.02  0.06  0.01 -0.06  0.00  0.00
##      [,26] [,27] [,28] [,29] [,30] [,31] [,32] [,33] [,34] [,35]
## ACF -0.07 -0.03 -0.01  0.04  0.05  0.01 -0.04  0.02  0.01 -0.01
## PACF -0.07  0.02  0.02  0.03  0.02 -0.04 -0.03  0.06 -0.03 -0.02
```

```
maxAR<-4+1
maxMA<-3+1
fits_all<-vector("list",length=maxAR)
AICmin<-BICmin<-Inf
for (ii in 1:maxAR){
fits_all[[ii]]<-vector("list",length=maxMA)
for (jj in 1:maxMA){
fits_all[[ii]][[jj]]<-invisible(sarima(design_matrix[, 1],p=ii-1,d=0,q=jj-1,xreg = design_
if(fits_all[[ii]][[jj]]$ICs["AICc"]<AICmin) AICmin<-fits_all[[ii]][[jj]]$ICs["AICc"]
if(fits_all[[ii]][[jj]]$ICs["BIC"]<BICmin) BICmin<-fits_all[[ii]][[jj]]$ICs["BIC"]
}
}
}
```

```
## <><><><><><><><><><><>
##
```

```

## Coefficients:
##           Estimate      SE t.value p.value
## intercept -10.9774 8.0102 -1.3704 0.1711
## xreg       1.3034 0.0398 32.7212 0.0000
##
## sigma^2 estimated as 13.53643 on 586 degrees of freedom
##
## AIC = 5.453466 AICc = 5.453501 BIC = 5.475796
##
## <><><><><><><><><><><><><><>
##
## Coefficients:
##           Estimate      SE t.value p.value
## ma1          0.1298 0.0296 4.3774 0.0000
## intercept   -8.7905 8.3231 -1.0562 0.2913
## xreg        1.2926 0.0414 31.2296 0.0000
##
## sigma^2 estimated as 13.11414 on 585 degrees of freedom
##
## AIC = 5.425202 AICc = 5.425272 BIC = 5.454976
##
## <><><><><><><><><><><><><><>
##
## Coefficients:
##           Estimate      SE t.value p.value
## ma1          0.1878 0.0347 5.4136 0.0000
## ma2          0.5652 0.0329 17.1608 0.0000
## intercept   -6.5233 5.9041 -1.1049 0.2697
## xreg        1.2813 0.0293 43.6641 0.0000
##
## sigma^2 estimated as 9.707088 on 584 degrees of freedom
##
## AIC = 5.129074 AICc = 5.12919 BIC = 5.166291
##
## <><><><><><><><><><><><><><>
##
## Coefficients:
##           Estimate      SE t.value p.value
## ma1          0.1984 0.0403 4.9210 0.0000
## ma2          0.5693 0.0339 16.7960 0.0000
## ma3          0.0209 0.0398 0.5257 0.5993
## intercept   -6.3751 5.8813 -1.0840 0.2788
## xreg        1.2805 0.0292 43.8099 0.0000
##
## sigma^2 estimated as 9.702543 on 583 degrees of freedom
##
## AIC = 5.13201 AICc = 5.132185 BIC = 5.17667
##
## <><><><><><><><><><><><><><>
##
## Coefficients:
##           Estimate      SE t.value p.value
## ar1          0.2423 0.0402 6.0324 0.0000
## intercept   -6.6371 8.4423 -0.7862 0.4321

```

```

## xreg          1.2819 0.0420 30.5357  0.0000
##
## sigma^2 estimated as 12.7465 on 585 degrees of freedom
##
## AIC = 5.396842  AICc = 5.396912  BIC = 5.426616
##
## <><><><><><><><><><>
##
## Coefficients:
##           Estimate      SE t.value p.value
## ar1          0.7079 0.0539 13.1325  0.0000
## ma1         -0.4520 0.0618 -7.3135  0.0000
## intercept   -6.8088 7.6773 -0.8869  0.3755
## xreg         1.2828 0.0382 33.6136  0.0000
##
## sigma^2 estimated as 11.97225 on 584 degrees of freedom
##
## AIC = 5.337735  AICc = 5.337852  BIC = 5.374952
##
## <><><><><><><><><><>
##
## Coefficients:
##           Estimate      SE t.value p.value
## ar1          0.0433 0.0760  0.5691  0.5695
## ma1          0.1571 0.0642  2.4470  0.0147
## ma2          0.5617 0.0340 16.5259  0.0000
## intercept   -6.3605 5.8764 -1.0824  0.2795
## xreg         1.2805 0.0292 43.8447  0.0000
##
## sigma^2 estimated as 9.701758 on 583 degrees of freedom
##
## AIC = 5.131929  AICc = 5.132104  BIC = 5.176589
##
## <><><><><><><><><><>
##
## Coefficients:
##           Estimate      SE t.value p.value
## ar1          0.5225 0.4056  1.2883  0.1982
## ma1         -0.3176 0.4108 -0.7732  0.4397
## ma2          0.4739 0.0850  5.5754  0.0000
## ma3         -0.2626 0.2396 -1.0959  0.2736
## intercept   -6.3970 5.8437 -1.0947  0.2741
## xreg         1.2807 0.0290 44.0991  0.0000
##
## sigma^2 estimated as 9.693055 on 582 degrees of freedom
##
## AIC = 5.134436  AICc = 5.134682  BIC = 5.18654
##
## <><><><><><><><><><>
##
## Coefficients:
##           Estimate      SE t.value p.value
## ar1          0.1414 0.0374  3.7752  0.0002
## ar2          0.4165 0.0374 11.1264  0.0000

```

```

## intercept  -8.5697 6.2535 -1.3704  0.1711
## xreg       1.2915 0.0311 41.5728  0.0000
##
## sigma^2 estimated as 10.52809 on 584 degrees of freedom
##
## AIC = 5.209682  AICc = 5.209799  BIC = 5.246899
##
## <><><><><><><><><><><><><><>
##
## Coefficients:
##           Estimate      SE t.value p.value
## ar1         -0.0189 0.0632 -0.2986  0.7653
## ar2          0.4618 0.0374 12.3497  0.0000
## ma1          0.1947 0.0647  3.0089  0.0027
## intercept   -7.9598 6.3701 -1.2496  0.2120
## xreg         1.2885 0.0317 40.7082  0.0000
##
## sigma^2 estimated as 10.37764 on 583 degrees of freedom
##
## AIC = 5.198749  AICc = 5.198925  BIC = 5.24341
##
## <><><><><><><><><><><><><><>
##
## Coefficients:
##           Estimate      SE t.value p.value
## ar1          0.0247 0.0738  0.3342  0.7384
## ar2          0.0827 0.0738  1.1200  0.2632
## ma1          0.1745 0.0640  2.7250  0.0066
## ma2          0.5071 0.0624  8.1280  0.0000
## intercept   -6.6067 5.8313 -1.1330  0.2577
## xreg         1.2817 0.0290 44.2292  0.0000
##
## sigma^2 estimated as 9.681434 on 582 degrees of freedom
##
## AIC = 5.133241  AICc = 5.133486  BIC = 5.185345
##
## <><><><><><><><><><><><><><>
##
## Coefficients:
##           Estimate      SE t.value p.value
## ar1          0.0896 0.6804  0.1318  0.8952
## ar2          0.0802 0.0808  0.9928  0.3212
## ma1          0.1094 0.6809  0.1606  0.8724
## ma2          0.4966 0.1222  4.0630  0.0001
## ma3         -0.0366 0.3825 -0.0957  0.9238
## intercept   -6.6254 5.8340 -1.1357  0.2566
## xreg         1.2818 0.0290 44.2114  0.0000
##
## sigma^2 estimated as 9.681359 on 581 degrees of freedom
##
## AIC = 5.136634  AICc = 5.136963  BIC = 5.196182
##
## <><><><><><><><><><><><><><>
##

```

```

## Coefficients:
##           Estimate      SE t.value p.value
## ar1          0.2122 0.0406  5.2208  0.0000
## ar2          0.4396 0.0373 11.7842  0.0000
## ar3         -0.1697 0.0408 -4.1599  0.0000
## intercept   -7.2640 6.4459 -1.1269  0.2602
## xreg         1.2850 0.0320 40.1115  0.0000
##
## sigma^2 estimated as 10.22559 on 583 degrees of freedom
##
## AIC = 5.184075  AICc = 5.184251  BIC = 5.228736
##
## <><><><><><><><><><><><><><>
##
## Coefficients:
##           Estimate      SE t.value p.value
## ar1          0.6328 0.1260  5.0207  0.0000
## ar2          0.3842 0.0477  8.0542  0.0000
## ar3         -0.3616 0.0552 -6.5474  0.0000
## ma1         -0.4310 0.1312 -3.2856  0.0011
## intercept   -7.6504 6.1739 -1.2391  0.2158
## xreg         1.2869 0.0307 41.9340  0.0000
##
## sigma^2 estimated as 10.06456 on 582 degrees of freedom
##
## AIC = 5.171702  AICc = 5.171948  BIC = 5.223806
##
## <><><><><><><><><><><><><><>
##
## Coefficients:
##           Estimate      SE t.value p.value
## ar1          0.0160 0.1254  0.1272  0.8988
## ar2          0.0819 0.0743  1.1028  0.2706
## ar3          0.0070 0.0796  0.0884  0.9296
## ma1          0.1830 0.1183  1.5474  0.1223
## ma2          0.5095 0.0676  7.5383  0.0000
## intercept   -6.6136 5.8371 -1.1330  0.2577
## xreg         1.2818 0.0290 44.1867  0.0000
##
## sigma^2 estimated as 9.681306 on 581 degrees of freedom
##
## AIC = 5.136629  AICc = 5.136957  BIC = 5.196176
##
## Warning in sqrt(diag(fitit$var.coef)): NaNs produced
## Warning in sqrt(diag(fitit$var.coef)): NaNs produced
##
## <><><><><><><><><><><><><><>
##
## Coefficients:
##           Estimate      SE t.value p.value
## ar1          0.3529   NaN    NaN    NaN
## ar2          0.0752 0.0805  0.9347  0.3503
## ar3         -0.0304   NaN    NaN    NaN
## ma1         -0.1537   NaN    NaN    NaN

```

```

## ma2          0.4493    NaN    NaN    NaN
## ma3          -0.1641    NaN    NaN    NaN
## intercept    -6.5983  5.8311 -1.1316  0.2583
## xreg         1.2817  0.0290 44.2291  0.0000
##
## sigma^2 estimated as 9.681391 on 580 degrees of freedom
##
## AIC = 5.140039  AICc = 5.140462  BIC = 5.20703
##
## <><><><><><><><><><><>
##
## Coefficients:
##           Estimate      SE t.value p.value
## ar1          0.1802  0.0405  4.4510  0.0000
## ar2          0.5223  0.0408 12.8003  0.0000
## ar3         -0.1299  0.0409 -3.1722  0.0016
## ar4         -0.1875  0.0406 -4.6163  0.0000
## intercept    -7.8300  5.8159 -1.3463  0.1787
## xreg         1.2878  0.0289 44.5484  0.0000
##
## sigma^2 estimated as 9.865657 on 582 degrees of freedom
##
## AIC = 5.151887  AICc = 5.152133  BIC = 5.203991
##
## <><><><><><><><><><><>
##
## Coefficients:
##           Estimate      SE t.value p.value
## ar1         -0.1028  0.1391 -0.7393  0.4600
## ar2          0.5852  0.0473 12.3818  0.0000
## ar3         -0.0045  0.0702 -0.0640  0.9490
## ar4         -0.2423  0.0426 -5.6824  0.0000
## ma1          0.2940  0.1404  2.0934  0.0367
## intercept    -7.7379  5.8667 -1.3189  0.1877
## xreg         1.2873  0.0292 44.1482  0.0000
##
## sigma^2 estimated as 9.803367 on 581 degrees of freedom
##
## AIC = 5.149005  AICc = 5.149333  BIC = 5.208552
##
## <><><><><><><><><><><>
##
## Coefficients:
##           Estimate      SE t.value p.value
## ar1         -0.0080  0.1449 -0.0551  0.9561
## ar2          0.1247  0.1532  0.8138  0.4161
## ar3          0.0132  0.0804  0.1639  0.8699
## ar4         -0.0303  0.0930 -0.3264  0.7442
## ma1          0.2070  0.1395  1.4836  0.1384
## ma2          0.4720  0.1371  3.4417  0.0006
## intercept    -6.7109  5.8375 -1.1496  0.2508
## xreg         1.2822  0.0290 44.1990  0.0000
##
## sigma^2 estimated as 9.679584 on 580 degrees of freedom

```



```
##
## AIC = 5.13985  AICc = 5.140273  BIC = 5.206841
##
## <><><><><><><><><><><>
##
## Coefficients:
##           Estimate      SE t.value p.value
## ar1          -0.9625 0.1253 -7.6802 0.0000
## ar2           0.1220 0.1451  0.8410 0.4007
## ar3           0.0900 0.1005  0.8960 0.3707
## ar4          -0.0128 0.0803 -0.1598 0.8731
## ma1           1.1709 0.1184  9.8856 0.0000
## ma2           0.6659 0.1564  4.2579 0.0000
## ma3           0.4950 0.0704  7.0309 0.0000
## intercept    -6.5961 5.8355 -1.1303 0.2588
## xreg          1.2817 0.0290 44.1947 0.0000
##
## sigma^2 estimated as 9.591682 on 579 degrees of freedom
##
## AIC = 5.137431  AICc = 5.137961  BIC = 5.211865
##
```

```
nn <- length(xx)
#look at AICs and BICs
print("check AICcs")
```

```
## [1] "check AICcs"
## [1] "check AICcs"
for (ii in 1:maxAR){
  for(jj in 1:maxMA){
    print(paste0(ii-1,"",jj-1))
    print(((fits_all[[ii]][[jj]])$ICs["AICc"] - AICmin) * nn)
  }
}
```

```
## [1] "00"
##      AICc
## 191.6673
## [1] "01"
##      AICc
## 174.9844
## [1] "02"
##      AICc
##      0
## [1] "03"
##      AICc
## 1.769839
## [1] "10"
##      AICc
## 158.2235
## [1] "11"
##      AICc
## 123.3187
## [1] "12"
```

```

##      AICc
## 1.722056
## [1] "13"
##      AICc
## 3.245475
## [1] "20"
##      AICc
## 47.63955
## [1] "21"
##      AICc
## 41.21284
## [1] "22"
##      AICc
## 2.538945
## [1] "23"
##      AICc
## 4.593384
## [1] "30"
##      AICc
## 32.54057
## [1] "31"
##      AICc
## 25.26943
## [1] "32"
##      AICc
## 4.590049
## [1] "33"
##      AICc
## 6.661523
## [1] "40"
##      AICc
## 13.55923
## [1] "41"
##      AICc
## 11.90428
## [1] "42"
##      AICc
## 6.549623
## [1] "43"
##      AICc
## 5.183149

print("check BICs")

## [1] "check BICs"

## [1] "check BICs"
for (ii in 1:maxAR){
  for(jj in 1:maxMA){
    print(paste0(ii-1,"",jj-1))
    print(((fits_all[[ii]][[jj]])$ICs["BIC"] - BICmin) * nn)
  }
}

## [1] "00"

```

```

##      BIC
## 182.9175
## [1] "01"
##      BIC
## 170.613
## [1] "02"
## BIC
##      0
## [1] "03"
##      BIC
## 6.134238
## [1] "10"
##      BIC
## 153.8521
## [1] "11"
##      BIC
## 123.3187
## [1] "12"
##      BIC
## 6.086455
## [1] "13"
##      BIC
## 11.96723
## [1] "20"
##      BIC
## 47.63955
## [1] "21"
##      BIC
## 45.57724
## [1] "22"
##      BIC
## 11.2607
## [1] "23"
##      BIC
## 17.66543
## [1] "30"
##      BIC
## 36.90497
## [1] "31"
##      BIC
## 33.99119
## [1] "32"
##      BIC
## 17.66209
## [1] "33"
##      BIC
## 24.07674
## [1] "40"
##      BIC
## 22.28099
## [1] "41"
##      BIC
## 24.97632
## [1] "42"

```

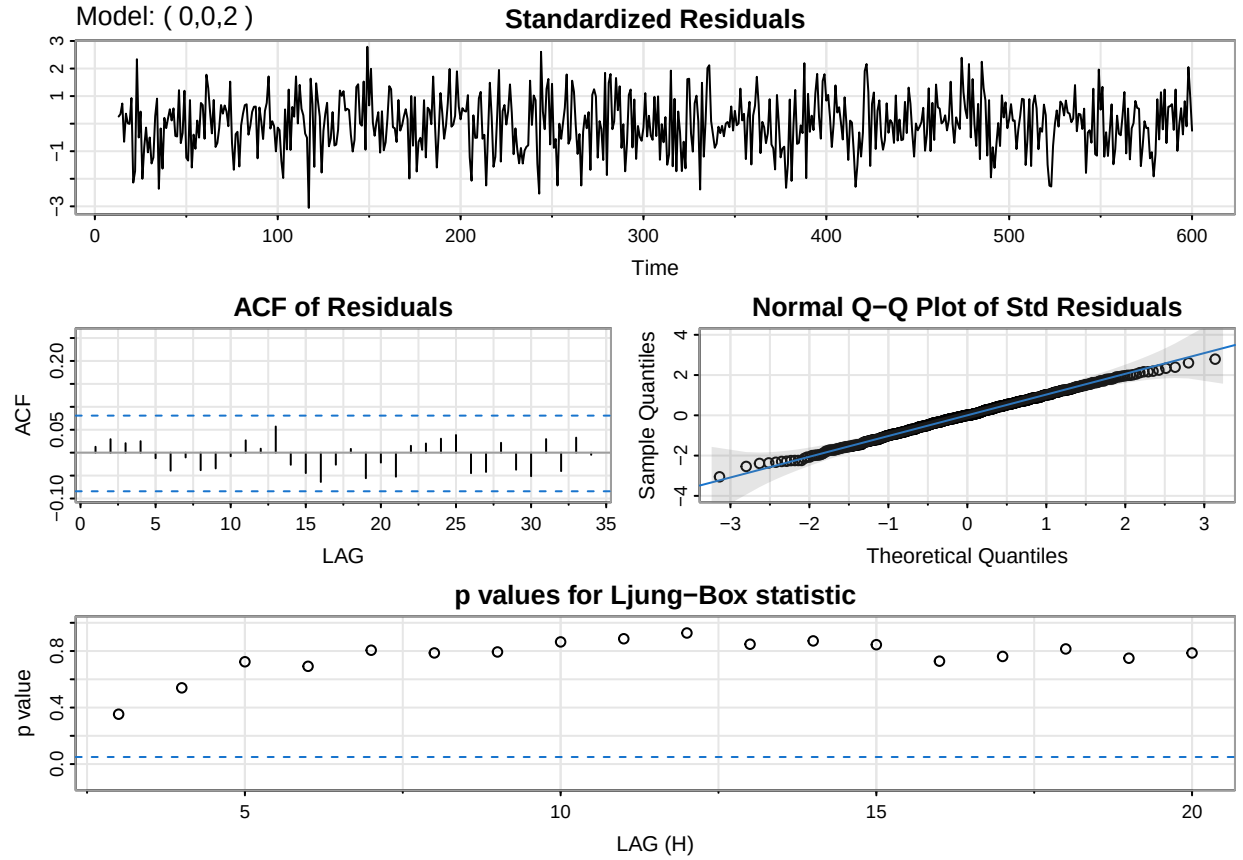
```

##      BIC
## 23.96484
## [1] "43"
##      BIC
## 26.93439

sarima(design_matrix[, 1],p=0,d=0,q=2,xreg = design_matrix[, 2],no.constant=FALSE, details=TRUE)

## initial  value 1.302692
## iter    2 value 1.148810
## iter    3 value 1.141258
## iter    4 value 1.136782
## iter    5 value 1.136525
## iter    6 value 1.136466
## iter    7 value 1.136432
## iter    7 value 1.136432
## iter    7 value 1.136432
## final   value 1.136432
## converged
## initial  value 1.137098
## iter    2 value 1.137096
## iter    3 value 1.137095
## iter    3 value 1.137095
## iter    3 value 1.137095
## final   value 1.137095
## converged
## <><><><><><><><><><><><><><>
##
## Coefficients:
##           Estimate      SE t.value p.value
## ma1           0.1878 0.0347  5.4136 0.0000
## ma2           0.5652 0.0329 17.1608 0.0000
## intercept    -6.5233 5.9041 -1.1049 0.2697
## xreg          1.2813 0.0293 43.6641 0.0000
##
## sigma^2 estimated as 9.707088 on 584 degrees of freedom
##
## AIC = 5.129074  AICc = 5.12919  BIC = 5.166291
##

```



Problem 2

Summary

The one dimensional DLM is of the form $Y_t = X_t + V_t$, $X_t = X_{t-1} + W_t$. The estimates generated for $\log(V_t)$ and $\log(W_t)$ are 2.285171 and 2.870191 respectively.

The two dimensional DLM is of the form $(Y_{t1}, Y_{t2}) = (1, 1)X_t + (V_{t1}, V_{t2})$, $X_t = X_{t-1} + W_t$. The estimates generated for $\log(V_{t1})$, $\log(V_{t2})$, and $\log(W_t)$ are 5.544694, 1.463094, and 1.617746 respectively.

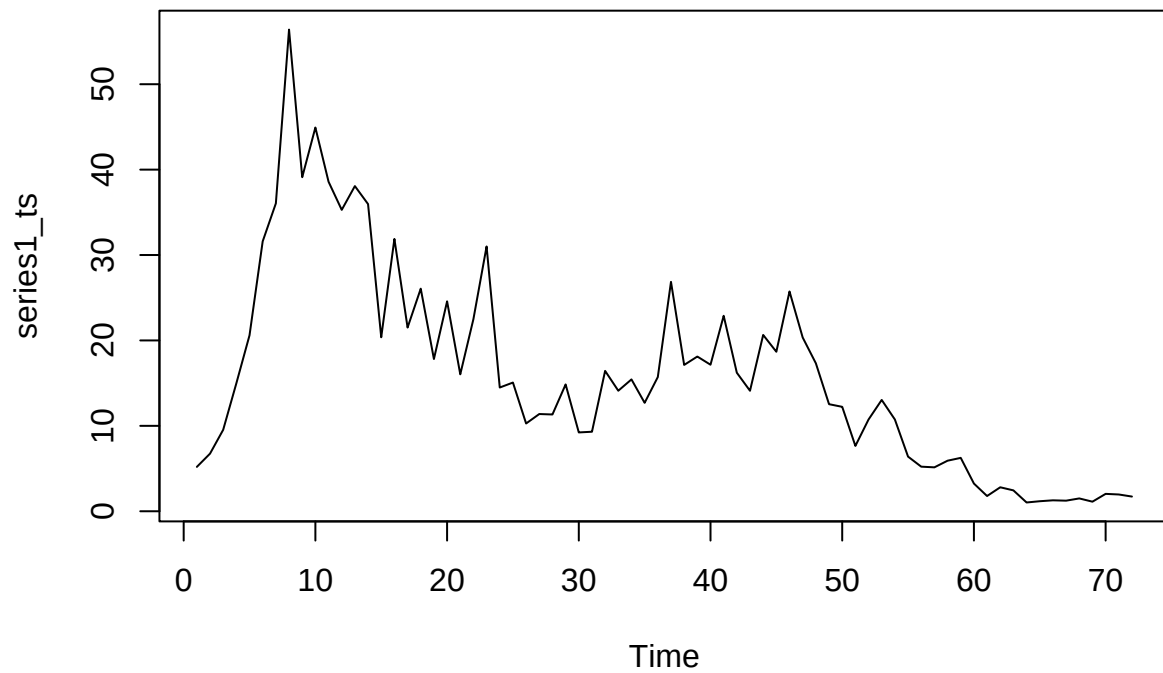
Explanation

- The estimates for $\log(V_t)$ and $\log(W_t)$ were generated using the `dlmMLE()` function for a one dimensional observation equation.
- The filter does not use future values of the series to predict current values, while the smoother does so. This means that the filter will fit the data more closely and will be more jagged than the smoother.
- The estimates for $\log(V_{t1})$, $\log(V_{t2})$, and $\log(W_t)$ were generated using the `dlmMLE()` function for a two dimensional observation equation.
- The filter and smoother for both data series track of both of the original series to some degree, while the filter and smoother for only the first series only tracks that series. In the two dimensional case, it is still the case that the smoother tracks the original series less strictly than the filter and is less jagged.

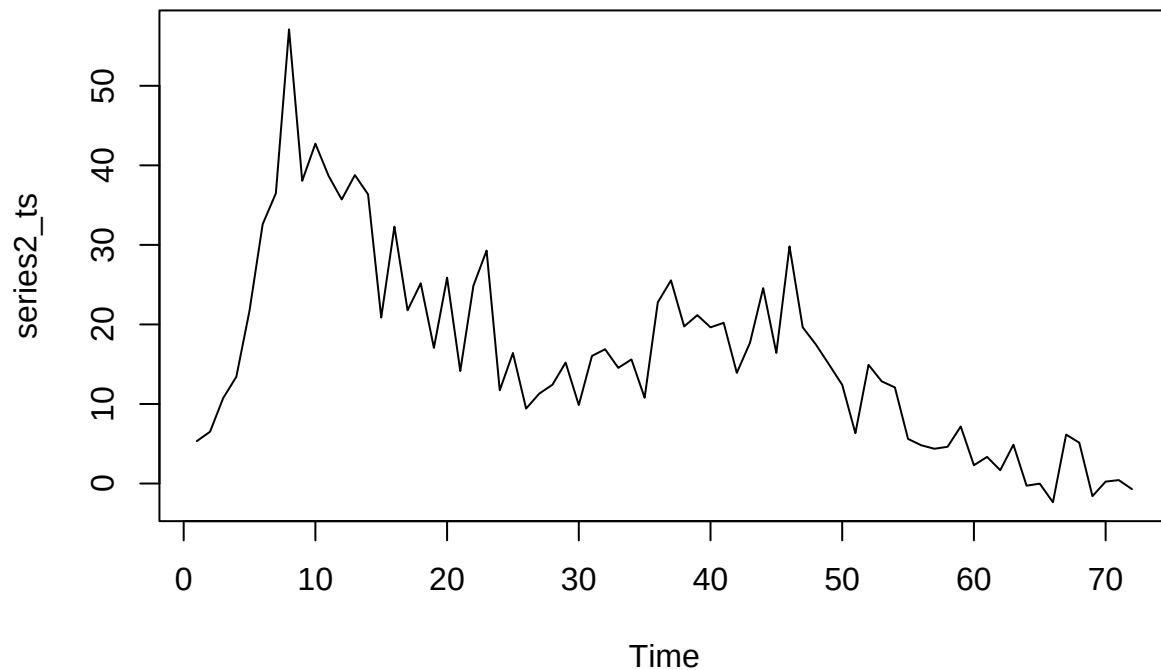
Output

```
# Convert third and fourth columns to time series objects
series1_ts <- ts(dat2[, 3])
series2_ts <- ts(dat2[, 4])

# Plot the time series
plot(series1_ts)
```



```
plot(series2_ts)
```



DLM parameters for one dimension (third column of original data) ($\log(V_t)$ and $\log(W_t)$ respectively):

```
library(dlm)
```

```
## Warning: package 'dlm' was built under R version 4.4.2
```

```
library(data.table)
```

```
library(ggplot2)
```

```
##
```

```
## Attaching package: 'ggplot2'
```

```
## The following object is masked from 'package:dlm':
```

```
##
```

```
##      %+%
```

```
mym0 <- series1_ts[1]
```

```
locallevel_1_1 <- function(x, m0){
```

```
  Sigma.R <- matrix(exp(x[1])) ## obs
```

```
  ## Sigma.Q <- diag(c(exp(x[3]), exp(x[4]))) ## state
```

```
  Sigma.Q <- matrix(exp(x[2]))
```

```
  m0 <- m0
```

```
  ## C0 <- diag(c(0,0))
```

```
  C0 <- matrix(.2^2)
```

```
  ## GG <- diag(c(1, 1)) ## state
```

```
  GG <- matrix(1) ## diag(1);
```

```
  FF <- matrix(1) ## diag(c(1))
```

```
  return(list(
```

```

    m0=m0,
    C0=C0,
    FF=FF,
    GG=GG,
    V=Sigma.R,
    W=Sigma.Q))}
dlmres <- dlmMLE(y = series1_ts,
                 parm = c(0,0),
                 m0 = mym0,
                 control = list(maxit=300),
                 build = locallevel_1_1)

print(dlmres$par)

```

```
## [1] 2.285171 2.870191
```

Filter and smoother for one dimensional DLM:

Black = original series Blue = filter Red = smoother

```

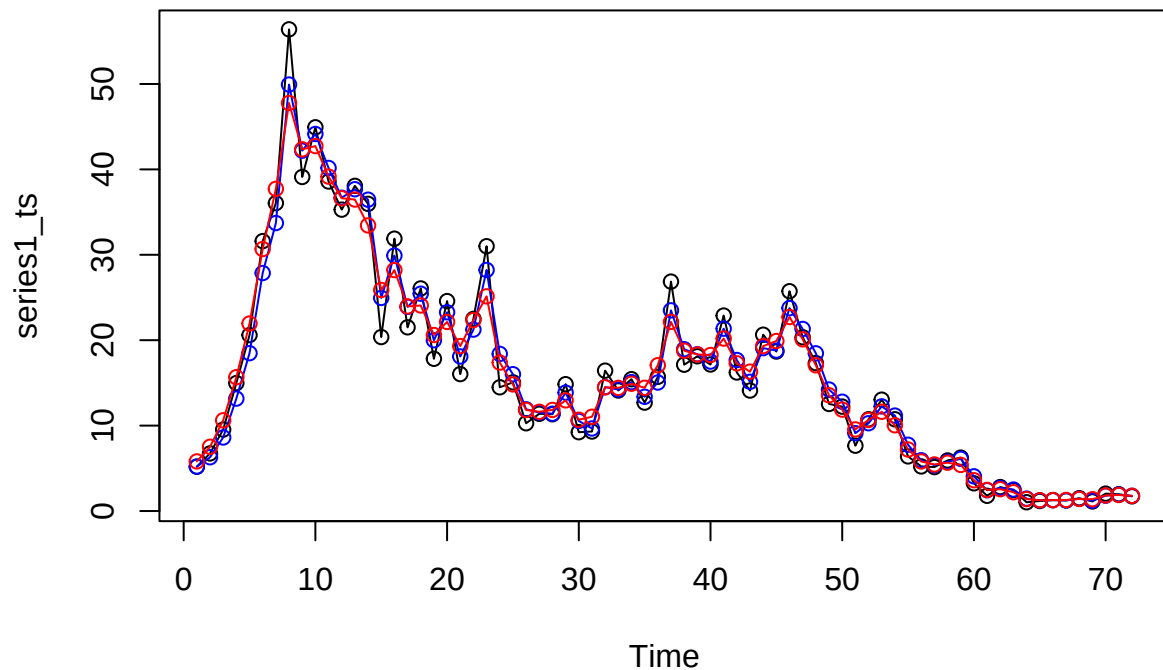
mod <- locallevel_1_1(x=dlmres$par, m0=mym0)
res_filter <- dlmFilter( y = series1_ts, mod = mod)

res_smoother <- dlmSmooth(res_filter)

# Convert filtered and smoothed results into time series
filter <- res_filter$m[-1]
filter_ts <- ts(data = filter)
smoother <- res_smoother$s[-1]
smoother_ts <- ts(data = smoother)

plot(series1_ts,
     type="o")
lines(filter_ts, type="o", col="blue")
lines(smoother_ts, type="o", col="red")

```

DLM parameters for two dimensions (third and fourth columns of original data) ($\log(V_{t1})$, ($\log(V_{t2})$, and $\log(W_t)$ respectively):

```
both_ts <- rbind(series1_ts, series2_ts)
mym02 <- matrix(mean(series1_ts))

locallevel_2_1 <- function(x, m0){
  Sigma.R <- diag(c(exp(x[1]), exp(x[2]))) ## obs
  ## Sigma.Q <- diag(c(exp(x[3]), exp(x[4]))) ## state
  Sigma.Q <- matrix(exp(x[3]))
  m0 <- as.matrix(m0)
  ## C0 <- diag(c(0,0))
  C0 <- matrix(data=.1^2)
  ## GG <- diag(c(1, 1)) ## state
  GG <- matrix(data=1)
  FF <- matrix(c(1,1), ncol=1)
  return(list(
    m0=m0,
    C0=C0,
    FF=FF,
    GG=GG,
    V=Sigma.R,
    W=Sigma.Q))}

dlmres_2 <- dlmMLE(y = (both_ts),
  parm = c(0,0,0),
  m0 = mym02,
```

```

        control = list(maxit=300),
        build = locallevel_2_1)
print(dlmres_2$par)

## [1] 5.544694 1.463094 1.617746
Filter and smoother for two dimensional DLM:
Black = first series Gray = second series Blue = filter Red = smoother
(mod2 <- locallevel_2_1(x = dlmres_2$par, m0=(mym02)))

## $m0
##      [,1]
## [1,] 15.96536
##
## $C0
##      [,1]
## [1,] 0.01
##
## $FF
##      [,1]
## [1,] 1
## [2,] 1
##
## $GG
##      [,1]
## [1,] 1
##
## $V
##      [,1]      [,2]
## [1,] 255.8764 0.000000
## [2,] 0.0000 4.319303
##
## $W
##      [,1]
## [1,] 5.041712

res_filter2 <- dlmFilter(y = t(both_ts), mod = mod2)

res_smoother2 <- dlmSmoother(res_filter2)

## output filter values ; first one has to be dropped
f2 <- ts(res_filter2$m[-1])
s2 <- ts(res_smoother2$s[-1])

## Plots
plot(series1_ts,
      type="o")
lines(series2_ts, type="o", col="gray" )
lines(f2, type="o", col="blue")
lines(s2, type="o", col="red")

```

